General Mathematics — Exercise Sheet 18

The complete solution set

Variant: Statements

Exercise 1 (Compatibility before translation). For $c \in \mathbb{R}$, consider the real system

$$x_1 + x_2 + x_3 = 2, \qquad x_1 - x_2 = 0, \qquad 2x_1 + x_3 = c.$$

Find all consistent values of c. For those values, find a particular solution p, a basis of the homogeneous solution space and the full family $p + \ker A$, where A is the coefficient matrix. State its rank and the null-space dimension convention of Lecture 18. Explain why its affine solution set is not a linear subspace.

Exercise 2 (Coordinates in a basis of the whole space). The columns

$$u_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad u_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad u_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

are real. Prove by solving the coefficient equations that their ordered list is a basis of $\mathbb{R}^3$. Find the coordinates of $w = (3, -2, 5)$ in this basis. Check the resulting linear combination componentwise. Use elimination and the basis definition, without determinants.

Exercise 3 (Two constraint sets). Let

$$V = \{(x, y, z, w) \in \mathbb{R}^4 : x + y = 0,\ z - w = 0\}, \qquad S = \{(x, y, z, w) \in \mathbb{R}^4 : x + y = 1,\ z - w = 0\}.$$

Find a basis of V and a column p such that $S = p + V$. Prove that $V \cap S = \varnothing$. For an arbitrary $q \in \mathbb{R}^4$, determine exactly when $q + V = S$. Justify both directions of your condition.

Exercise 4 (The scalar field controls the parameters). Find every complex solution of

$$x_1 + \mathrm{i}x_2 = 1, \qquad x_2 + \mathrm{i}x_3 = 0.$$

Give a particular solution and an ordered basis of its null space over $\mathbb{C}$. State the rank and null-space dimension convention over that field. Would allowing only real values of the free parameter give every complex solution? Give an explicit test column supporting your answer.

Exercise 5 (Zero and empty-basis cases). Let $K = \mathbb{R}$ or $\mathbb{C}$ and consider the two coefficient matrices

$$Z = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

For each matrix, find its rank, a basis of its null space and that null space's dimension under the lecture's convention. Classify all right sides for which $Zx = b$, $b \in K^2$, is consistent, and all $(\alpha, \beta, \gamma) \in K^3$ for which $By = (\alpha, \beta, \gamma)$ is consistent. Give all solutions in every consistent case. Explain the difference between an empty solution set and a null space with an empty basis.